

Scene Setup Guide

MainMenu Scene Setup

Hierarchy Structure:

```

MainMenu
├── Main Camera
├── Canvas
│   ├── MainMenuPanel
│   │   ├── TitleText
│   │   ├── PlayButton
│   │   ├── OptionsButton
│   │   ├── CreditsButton
│   │   ├── QuitButton
│   │   └── HighScoreText
│   ├── OptionsPanel
│   │   ├── OptionsTitle
│   │   ├── VolumeSlider
│   │   ├── FullscreenToggle
│   │   └── BackButton
│   └── CreditsPanel
│       ├── CreditsTitle
│       ├── CreditsText
│       └── BackButton
├── EventSystem
└── MenuManager
  
```

Setup Steps:

1. **Create Scene:** File > New Scene, save as "MainMenu"
2. **Camera Setup:**
 - Set Clear Flags to "Solid Color"
 - Set Background to dark blue/black (space theme)
3. **Canvas Setup:**
 - Add Canvas (UI > Canvas)
 - Set UI Scale Mode to "Scale With Screen Size"
 - Reference Resolution: 1920x1080
4. **MainMenuPanel:**
 - Add Panel, name "MainMenuPanel"
 - Add TitleText (TextMeshPro - Text): "SPACE SHOOTER"
 - Add Buttons for Play, Options, Credits, Quit
 - Add HighScoreText below buttons
5. **OptionsPanel:**
 - Create Panel, name "OptionsPanel", disable initially
 - Add Volume Slider
 - Add Fullscreen Toggle
 - Add Back Button

6. CreditsPanel:

- Create Panel, name "CreditsPanel", disable initially
- Add title and credits text
- Add Back Button

7. MenuManager:

- Create empty GameObject, add MenuManager script
- Assign all panel and UI references
- Connect button OnClick events

8. Add Background (optional):

- Add scrolling star background for visual appeal

GameScene Setup

Hierarchy Structure:

```

GameScene
├── Main Camera
├── Managers
│   ├── GameManager
│   ├── ScoreManager
│   ├── WaveManager
│   └── EnemySpawner
├── Player (spawned by GameManager)
├── Background
│   ├── ScrollingBackground
│   └── Stars (Parallax layers)
├── Canvas
│   ├── HUD
│   │   ├── HealthBar
│   │   ├── HealthText
│   │   ├── ScoreText
│   │   ├── HighScoreText
│   │   └── WaveText
│   ├── PausePanel
│   │   ├── PausedTitle
│   │   ├── ResumeButton
│   │   ├── RestartButton
│   │   ├── MainMenuButton
│   │   └── QuitButton
│   ├──GameOverPanel
│   │   ├──GameOverTitle
│   │   ├──FinalScoreText
│   │   ├──HighScoreText
│   │   ├──RestartButton
│   │   └──MainMenuButton
│   └──WaveCompletePanel
│       └──WaveCompleteText
├── EventSystem
└── PlayerSpawnPoint
  
```

Setup Steps:

1. **Create Scene:** File > New Scene, save as "GameScene"

2. Camera Setup:

- Position: (0, 0, -10)
- Orthographic Size: 5
- Background: Dark blue/black
- Add CameraShake script

3. Create Managers Object:

- Empty GameObject "Managers"
- Add children with respective scripts:
 - GameManager (with player prefab reference)
 - ScoreManager
 - WaveManager
 - EnemySpawner (configure enemy types list)

4. PlayerSpawnPoint:

- Empty GameObject at position (0, -3, 0)

5. Background Setup:

- Add ScrollingBackground prefab
- Or create parallax layers with ParallaxBackground script

6. Canvas & UI:

- Create Canvas with Scale Mode "Scale With Screen Size"
- Create HUD panel (always visible):
 - Health bar (Slider)
 - Score text (TextMeshPro)
 - Wave counter
 - Create PausePanel (disabled by default)
 - Create GameOverPanel (disabled by default)
 - Create WaveCompletePanel (disabled by default)

7. UIManager:

- Add UIManager script to Canvas or Managers
- Assign all UI references

8. Connect Everything:

- Assign prefabs to GameManager
- Assign enemy prefabs to EnemySpawner
- Connect button callbacks to UIManager methods
- Wire up HealthSystem.OnHealthChanged to UIManager.UpdateHealthBar

Build Settings

1. File > Build Settings
2. Add scenes in order:
 - MainMenu (index 0)
 - GameScene (index 1)
3. Select "PC, Mac & Linux Standalone"

4. Target Platform: Windows
5. Architecture: x86_64
6. Click "Build" to create the executable